



Credit that grows from every sale.

Modal usaha, dari hasil jualan sendiri.

AI micro-credit for Indonesia's small shops, enforced by a smart contract on BNB Chain.



AI Agents



Finance & Commerce



Consumer Apps

Indonesia Web3 Hackathon 2026 · @ezrahesperos

64 million small shops.

No record a lender trusts.

"Pinjol" is the Indonesian name for online lending apps, many of them predatory.



Sales live in a notebook.

The income is real, but cash and paper prove nothing to a lender.



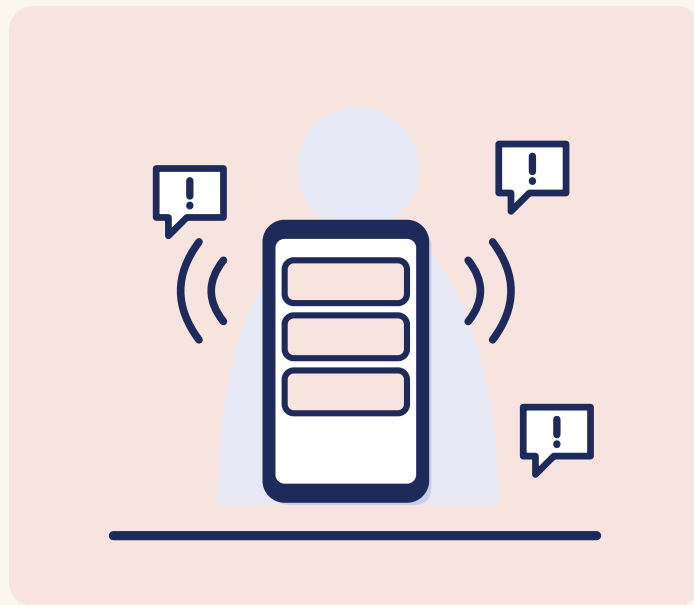
So they turn to pinjol.

Interest that keeps growing, collectors who call the family, apps that read their contacts.



No fair working capital.

A shop that sells every day cannot borrow to buy tomorrow's stock.



Every sale becomes credit.



No collateral



No collectors, no due date



Flat fee, written up front, at most 5%

Built for a shop owner, not for a crypto user.



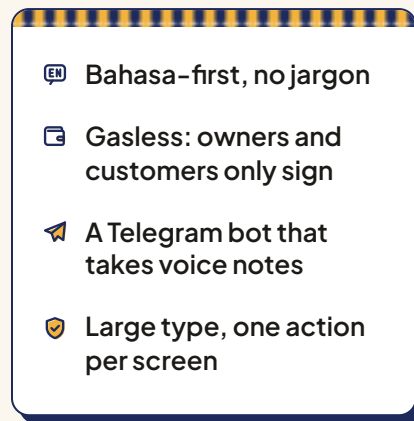
Show the QR. The customer pays in seconds.



See today's sales. And how much capital is possible.



Ask, even by voice note. In the app or on Telegram.



Screens from the live app. The interface is in Bahasa Indonesia.

The limit comes from real sales, not from a score.

Credit limit - Warung Bu Sri

Verified sales, last 30 days Rp 12.115.000

× 10% Rp 1.211.500

Level ceiling (Berkembang) Rp 2.000.000

Credit limit **Rp 1.211.500**

The smaller of the two, computed by the smart contract.



Each repaid loan raises the level, and the ceiling doubles.

 At least 5 different customers

 One customer counts for at most Rp 250.000 a day

Paying yourself, or one friend paying a lot, does not raise the limit.

Plain terms. Repaid from sales. No due date.

Loan offer

You receive **Rp 800.000**

Flat fee **Rp 24.000**

Total to repay Rp 824.000

Taken from each sale **10%**

- ✔ No due date
- ✔ No late fees
- ✔ No collectors

Terms are read from the contract, not from the chat.
Accepting takes one signature.

One sale

Rp 50.000

Rp 45.000 to the shop

Rp 5.000 repays the loan

Each sale splits inside the payment.



LUNAS!

Repaid in full.



✔ **Level up: Berkembang**

New ceiling: Rp 2 jt.

A slow week only means slower repayment. This exact loan was made and repaid on-chain.

The AI proposes. **The contract decides.**



- 1 The AI's key can only propose.**
It holds no funds; accepting needs the owner's signature.
- 2 Hard caps in the contract.**
Limit, fee at most 5%, repayment at most 20%, pool exposure, daily budget.
- 3 The AI never writes numbers.**
Code fills every figure from the chain.
- 4 Reply guard.**
A figure not found in a tool result forces a rewrite.

We attacked it.

Request: Rp 1.000.000.000 at 0% fee



Blocked by the policy layer: above the shop's ceiling.

Same request, policy layer switched off

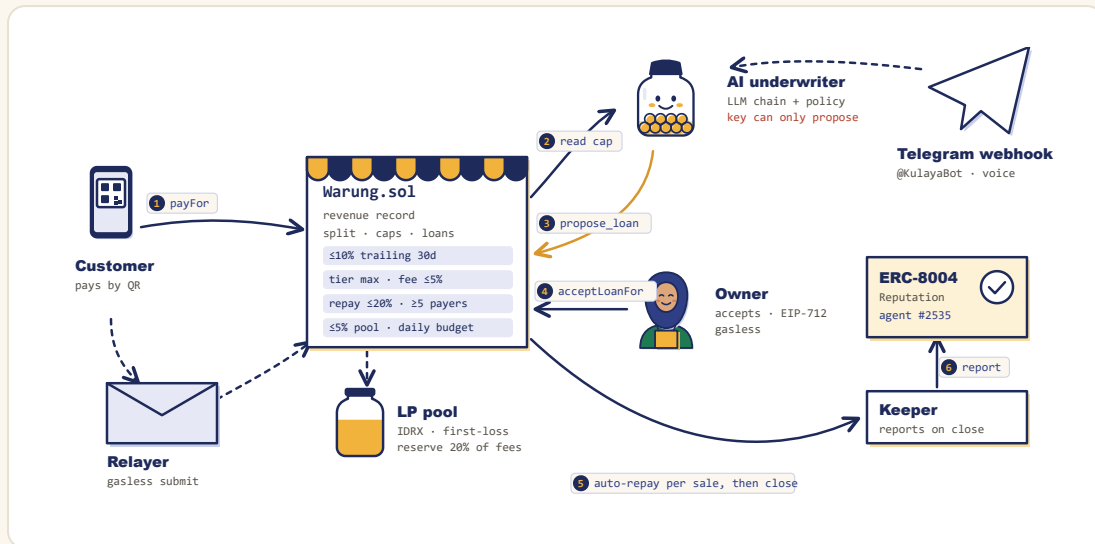


The contract alone still rejects it.

SIMULATION · NOTHING IS BROADCAST

Try it: kulaya.vercel.app/protocol/redteam

What a bank can't see, the chain can prove.



- A record the shop owns.**
Every sale is a public event any lender can verify.
- Repayment by code.**
The split happens inside the payment; no collector needed.
- An open pool.**
Anyone can fund shops; 20% of every fee goes to a first-loss reserve.
- An AI with a track record.**
The underwriter is ERC-8004 agent #2535:
100/100 from 1 loan

Gasless: users sign EIP-712 messages, a relayer submits them, and the contract moves only exactly what was signed.

Working on testnet. Here is what's real, and what isn't yet.

Built and verified

- 🧐 Smart contracts with **33** Foundry tests: unit, fuzz, invariants, relay attacks
- 🧐 App, AI agent and bot with **32** unit tests, plus live end-to-end browser journeys
- 🧐 A full loan cycle on-chain: proposed, accepted, repaid by **50** customer payments, closed
- 🧐 The AI's on-chain reputation: **100/100** from **1** loan
- 🧐 **30** shops registered on testnet (demo and test shops)

Honest limits

- ⚠️ Testnet only. The rupiah stablecoin is a mock (MockIDRX); no real money.
- ⚠️ The QR is not QRIS yet. OVO, GoPay and bank apps cannot scan it.
- ⚠️ No real pilot users yet.
- ⚠️ A real launch needs a licensed lender of record.

Verify it: contract **0xF6fD...21D8** on BNB Smart Chain Testnet (chain 97).

Next: real rupiah, through QRIS.

QRIS bridge.

Customers pay with any e-wallet; a licensed provider settles IDR into the same contract.

Licensed lending partner.

A licensed P2P lender or koperasi as lender of record, through the OJK sandbox.

Mainnet with real IDR.

Invisible wallets.

Embedded wallets and a paymaster, so owners never see crypto.



Try the app
kulaya.vercel.app



Try to break it
[/protocol/redteam](https://protocol/redteam)



Watch the demo
youtu.be/mTLAuLq-RCg



Telegram: @KulayaBot

Code: github.com/ketutezraugm/kulaya

Kulaya: credit that grows from every sale.

Built solo by @ezrahesperos for Indonesia Web3 Hackathon 2026.